

Experience Is Meaning-Giving

A Strong-Form Readout-Retention Theory of Phenomena, Resonance, Rhythm, Accumulation, and Transformative Release

A programme synthesis from Readout Genesis, Human LoRA, Human Learning, Meaning Before Naming, Fusion/Tunnel, and CTSA Human Return

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Status. K0 strong-form conceptual and measurement architecture, revised after adversarial peer review without withdrawing the paper's central claims. The identity thesis is retained: for a finite situated human, experience is the giving of meaning to a phenomenon. Revision strengthens rather than narrows that claim by separating distinct modes of significance, exposing direct rival explanations, and specifying what each construct must add empirically. Resonance remains external-internal experiential congruence; rhythm remains structured recurrence across ordered history; accumulation, informational work, barrier crossing, and transformative release remain explicit candidate transition structures. None is literal physical energy or an automatic truth certificate. Internal programme manuscripts establish genealogy; external literature constrains novelty and identifies rival mechanisms.

Central claim. A phenomenon becomes an experience for a finite human by being given meaning: it becomes affectively, pragmatically, autobiographically, conceptually, or epistemically significant within a reader-conditioned relation to embodiment, retained history, expectation, value, available concepts, and context. These are modes of one meaning-giving relation, not competing definitions of experience. Meaning-giving can begin before stable naming, while naming can recursively reorganize later experience. When a present externally prompted experience corresponds with retained internal experiential organization, **resonance** occurs as external-internal congruence. When encounters recur with structured timing, **rhythm** organizes recurrence, spacing, interruption, emphasis, and return. Repeated retained changes can accumulate transition readiness; sufficiently aligned encounters may contribute to barrier crossing and **transformative release**. None of these transitions certifies truth, healing, wisdom, or improvement. In Human-AI interaction, the prompt is therefore not the beginning of the human trajectory but a bounded articulation of an already meaning-shaped state.

1. Abstract

What is an experience? The Human-AI Readout Programme has progressively separated world events, retained traces, meaning, memory, knowledge, language, and human return, but its earlier manuscripts stopped short of a single defended theory of how a phenomenon becomes lived experience. This paper proposes that *experience is meaning-giving*: not the arbitrary projection of a private label onto neutral data, and not the reception of a finished semantic object already contained in the world, but a finite, embodied, affectively weighted, history- and context-conditioned readout relation through which a phenomenon becomes significant for a particular agent. The proposal is rooted internally in Readout Genesis/MEMK, where meaning follows readout and experience is formally represented as an event-trace-meaning configuration; in *Experience Is the Human LoRA*, where experience is temporally thick and selected residues can alter future interpretation; in *Human Learning as Epistemic Architecture*, where lived words make experience inspectable and revisable; in *From Problem to Hypothesis* and *Fusion/Tunnel*, where accessibility is history-shaped and repeated traversal can alter future routes; and in *CTSA Human-Return Readout*, where retained change is distinguished from demonstrable unaided capability.

External literature constrains rather than vali-

dates the architecture. Phenomenology treats conscious experience as intentional and horizon-laden; enactivism frames cognition as sense-making in organism-environment coupling; appraisal theory distinguishes event appraisal from later feeling and labeling; emotion research treats experience as a content-rich binding of affect, perception, and conceptual knowledge; predictive-processing work shows how prior models can shape perception while recent reviews caution against over-generalization; memory and insight research supports selective retention, reconsolidation, and abrupt representational restructuring; music-cognition research establishes rhythm as a measurable temporal structure with competing mechanistic models; and Human-AI experiments show that repeated AI interaction can alter later human judgments.

The paper contributes a strong-form integrative architecture rather than merely a new label for significance. First, **meaning-giving** is typed as a plural but unified relation that can carry affective, pragmatic, autobiographical, conceptual, and epistemic significance without reducing experience to language. Second, **experiential resonance** names external-internal experiential congruence while remaining empirically separable from schema congruence, self-reference, familiarity, processing fluency, appraisal, and memory retrieval. Third, **rhythmic organization** names structured recurrence across ordered history and is connected not only to music

cognition but also to event segmentation, memory, learning, and dialogue. Fourth, **transformative release** connects accumulated retained change, transition readiness, barrier crossing, and abrupt reorganization while remaining distinguishable from mere intensity and from durable transformation. Fifth, the paper names the **Pre-Prompt Human State Principle**: Human-AI interaction begins from a history-shaped human state whose prior experiences already condition the next prompt. A formal architecture, non-collapse laws, rival-model tests, model-selection criteria, and legitimate defeaters make these claims empirically vulnerable without weakening their conceptual scope.

2. The Residual Problem: From Phenomenon to Experience

The programme already distinguishes a world-side occurrence from its representation and interpretation. Readout Genesis states that a retained difference precedes the name of what differs and that meaning is attached only after domain translation and readout [1]. In its MEMK domain, meaning is explicitly defined as a readout-conditioned relation rather than a property stored in the root, and experience is represented as a non-additive event-trace-meaning configuration [2]. This gives an immediate clue: a phenomenon is not yet identical to experience.

The central residual question is therefore not whether humans “have experiences.” It is what architecture connects an encountered phenomenon to a lived, significant event without collapsing one into the other. Three inadequate answers must be avoided.

First, **copying**: the experience is treated as a faithful internal copy of an external event. This ignores selective access, embodiment, prior learning, expectation, and context. Second, **semantic containment**: the phenomenon is treated as if it already carries the same finished meaning that every observer merely extracts. This fails wherever the same event matters differently to different readers. Third, **arbitrary construction**: meaning is treated as unconstrained private invention. This ignores source structure, common embodiment, social history, and world resistance.

The proposed alternative is relational:

A phenomenon becomes an experience when a finite situated agent gives it significance through a constrained readout relation between what is encountered and what the agent already retains, expects, values, can discriminate, and can act upon.

The phrase *gives meaning* therefore does not mean “makes truth.” It means that a phenomenon becomes *for this agent, in this context, under this history* something threatening, familiar, beautiful, actionable, puzzling, sacred, irrelevant, consoling, ab-

surd, or otherwise significant. The resulting significance may later be revised, rejected, or shown false.

3. Internal Genealogy and the New Residual Contribution

The present paper does not replace the programme’s earlier objects. It fixes their order and identifies a missing interface.

The new residual contribution is narrower than a universal theory of consciousness but broader than the earlier *Meaning Before Naming* note. It is a theory of **experience formation and retained transformation**:

phenomenon → meaning-giving readout
 → experience → retention/revision
 → future readout.

The paper then asks how resonance, rhythm, and transition belong inside this sequence.

4. External Literature as Constraint, Not Validation

The proposed architecture sits near several mature traditions. Its novelty claim survives only if it does not rename them as discoveries.

4.1 Phenomenology: experience is intentional and horizon-laden

Phenomenology has long treated conscious experience as directed: perception, memory, judgment, emotion, and imagination are experiences *of* or *about* something. Husserlian and Merleau-Pontian traditions also emphasize the lived body and the largely implicit horizon that gives an experienced object background significance [10, 11]. This is an important ally to the present thesis because it blocks the idea that experience is a bare sensory registration followed only later by optional meaning. Yet phenomenology alone does not supply the programme’s retention gates, provenance rules, Human-AI session boundary, or empirical falsifiers.

4.2 Enaction: meaning as sense-making in organism-environment coupling

Enactive theory comes even closer to the claim that experience involves meaning-giving. Di Paolo argues that adaptivity is required to ground sense-making and agency, while Thompson’s enactive synthesis treats significance and valence as arising in the living organism’s regulated coupling with its environment rather than being intrinsic labels waiting to be extracted [12, 13, 11]. The overlap is substantial: meaning is relational, embodied, history-sensitive, and action-relevant. The present paper does not claim to discover sense-making. Its narrower contribution is to place a readout-retention architecture around this relation, then connect it to resonance, rhythmic recurrence, retained transformation, and Human-AI feedback.

Table 1: Internal programme genealogy. Self-citation establishes lineage and role separation, not independent empirical validation.

Programme object	Retained result	Function here
Readout Genesis / MEMK	Retained difference precedes semantic naming; meaning follows readout; experience is an event-trace-meaning configuration; meaning, truth, and epistemic status are non-identical.	Supplies the level distinction between phenomenon, meaning-giving, experience, memory, and knowledge.
Experience Is the Human LoRA	Experience is temporally thick, embodied, affectively weighted, and context-mediated; selected residues may persist and alter future interpretation/action.	Supplies the retention and operator-change layer.
Human Learning as Epistemic Architecture	Lived words connect experience, emotion, value, memory, evidence, goals, constraints, problem-context, tools, workflow, skill, and feedback.	Supplies inspectable articulation and the concept-to-competence arc.
From Problem to Hypothesis	Accessibility differs from reachability; history-shaped attraction and recent-path momentum may change speed and direction of semantic mobility.	Supplies graded re-entry, path history, and auditable restructuring.
Epistemic Fusion or Tunnel	Human-AI dialogue recursively changes accessible routes; repeated traversal can create inclination; amplification is epistemically neutral; only retained human change crosses the session boundary.	Supplies reciprocal interaction, temporalized accessibility, resistance, and direction.
Meaning Before Naming	Pre-explicit affective-relational organization can precede stable naming; naming can later restructure the state it reports.	Supplies the pre-articulation interval but requires correction of the earlier resonance definition.
CTSA Human-Return Readout	After AI removal, retained change is classified by demonstrable Concept, Tool-Selection, Skill-Execution, or Alternative-Generation Return, with loss/warrant/direction kept separate.	Supplies the upper measurement boundary: explicit crystallized return, not every retained residue.

4.3 Appraisal and emotion construction: event meaning before stable label

Scherer and Moors review evidence for an emotion process in which appraisal of an event leads to differentiated action tendencies and physiological responses before conscious feeling and later semantic categorization or labeling [14]. Barrett and colleagues describe emotion experience as a content-rich event in which evolving affect, world perception, and conceptual knowledge are bound at a moment in time [15]. These literatures support a layered process from event significance to explicit category, while also warning that affect, conceptual knowledge, and labeling interact.

The warning is critical. Lindquist and colleagues experimentally reduced the accessibility of emotion words and impaired the speed or accuracy of emotion perception [16]. Lieberman and colleagues found that affect labeling changes neural and affective responding to negative stimuli [17]. Thus naming cannot be modeled as a passive final report. The strong one-way formula

meaning → name

is inadequate. Naming may reveal, compress, stabilize, distort, or actively reorganize what is experienced.

4.4 Predictive processing: prior organization shapes present perception

Predictive-processing accounts model perception as shaped by prior expectations and error correc-

tion rather than as bottom-up copying [18]. This strongly constrains any readout theory that would ignore prior organization. However, the current predictive-processing literature is neither conceptually nor empirically uniform. A 2026 Annual Review argues that sensory prediction-error findings can reflect different computations and that the framework's definitions and scope require renewed scrutiny [19]. The present paper therefore uses prediction and surprise as possible mechanisms or covariates, not as a universal explanatory foundation.

4.5 Memory, reconsolidation, and insight: retained history can be revised abruptly

Prior knowledge and schemas alter encoding, consolidation, and retrieval [20, 21]; emotional memory is selectively remembered and retrieval can itself alter later representation [22]. Reconsolidation research further supports the possibility that reactivated memories can become modifiable, though mechanistic and translational claims require care [23]. Insight research distinguishes abrupt representational restructuring from the phenomenological Aha! experience; an Aha! feeling and a genuine lower-defect restructuring need not be the same event [24]. These literatures constrain the present transition model: abrupt subjective release is possible, but intensity alone cannot identify the underlying mechanism, truth, or durability.

4.6 Rhythm: measurable temporal structure, not semantic substance

Research on musical rhythm, beat, and metre treats rhythm as a measurable organization of events in time and compares mechanistic accounts including oscillator and predictive-coding models [25]. A large cross-cultural study of 39 participant groups in 15 countries found both shared small-integer-ratio rhythm priors and systematic culture-related variation, showing that temporal structure can be jointly constrained by common regularities and learned history [26]. The lesson for the present architecture is exact: rhythm is not a metaphor for meaning and is not itself emotion. It is a temporal structure that can constrain prediction, coordination, expectation, and re-entry. The programme extends this idea beyond music only at the level of a testable domain hypothesis: recurrence, spacing, interruption, emphasis, and return may organize later accessibility in learning and dialogue.

4.7 Human-AI feedback: the human state can change before the next prompt

Glickman and Sharot report that repeated interaction with biased AI systems alters later human perceptual, emotional, and social judgments, producing feedback loops that can amplify bias [32]. This does not validate the full readout-retention theory, but it directly defeats the assumption that each prompt starts from a fixed human baseline. Prior interaction can alter the human state from which the next prompt is produced.

4.8 Schema congruence, self-reference, and fluency: direct rivals for resonance

The resonance construct is intentionally placed under stronger pressure than in the previous draft. Schema research shows that information congruent with prior knowledge is often encoded and consolidated differently from incongruent information [27]. A meta-analysis of the self-reference effect found robust memory advantages when material is related to the self, with organization and elaboration accounting for much of the effect [28]. Processing-fluency research further shows that ease of processing can increase liking and felt positivity without requiring a special external-internal congruence mechanism [29]. These are not reasons to abandon resonance; they define the rival set it must beat. The strong claim is that resonance, if retained as a technical construct, captures an experiential fit relation that cannot be exhausted by any one of schema congruence, self-reference, familiarity, fluency, appraisal, or retrieval alone.

4.9 Event segmentation: temporal organization beyond music

Event-cognition research provides the missing bridge from musical rhythm to temporally organized experience. Event Segmentation Theory proposes that people spontaneously partition ongoing

activity into meaningful events, with event boundaries linked to changes in prediction and with consequences for attention and memory [30]. Related work emphasizes that segmentation depends on both bottom-up perceptual change and top-down conceptual structure such as goals [31]. This literature does not prove a general rhythm law. It does show that the temporal form of experience is not exhausted by clock time or exposure count: boundaries, return, transition, and predictive discontinuity can reorganize how episodes are encoded and remembered.

5. The Defended Thesis: Experience Is Meaning-Giving

The title is intentionally strong. It should not be read as saying that human agents arbitrarily manufacture the truth of phenomena. The defended thesis is a relational identity at the level of lived experience:

For a finite situated human, an experience is the phenomenon as it becomes meaningfully present through a constrained relation among encountered structure, embodiment, retained history, expectation, value, conceptual resources, and context.

Three consequences follow.

P1 - Phenomenon is not experience. An occurrence or trace can exist without being experienced by a particular reader. Experience begins only when some part of that occurrence becomes accessible and significant to the reader.

P2 - Meaning-giving is not truth-making. The fact that an event is experienced as betrayal, beauty, threat, revelation, injustice, or opportunity establishes the reader's experienced significance, not the objective truth of every interpretation attached to it.

P3 - Meaning-giving is constrained. The reader is not free to assign any meaning whatsoever without cost. Source structure, embodiment, learned categories, social practice, memory, prediction, rights, and later world resistance constrain admissible and stable interpretations.

P4 - Experience can be pre-explicit. Some significance may be active before the person can stably name it, but this does not imply a fully formed proposition hidden beneath language.

P5 - Naming is recursive. Later articulation can reveal, sharpen, compress, distort, or reorganize the experience that made articulation possible.

P6 - No lived experience is significance-free. Under the present thesis, a phenomenon can be registered without becoming a lived experience for a declared human reader; but once it is lived as experience, at least one significance relation is active. That significance need not be lexical or propositional. Pain may be affectively and pragmatically significant before it is conceptually named; a face may be autobiographically salient before its relevance is

Table 2: External literature map and the residual role of the present paper.

Literature	What it already provides	Constraint / residual role here
Phenomenology	Intentionality, lived body, implicit horizon of meaning.	Do not claim that significance in experience is newly discovered; add formal level separation and retention.
Enaction / sense-making	Meaning and value emerge relationally in embodied organism-environment coupling.	Position meaning-giving as a readout-retention extension, not a substitute for enaction.
Appraisal / emotion construction	Event appraisal can precede labeling; affect, perception, and concepts jointly shape experience.	Pre-explicit organization is plausible, but language can also be constitutive.
Predictive processing	Prior models and prediction errors shape perception/action.	Use prediction/surprise as rival or component mechanisms, not as automatic validation.
Memory / reconsolidation / insight	Prior knowledge shapes encoding; retained memories can update; restructuring can be abrupt.	Separate release from durable transformation and Aha! from warranted insight.
Rhythm cognition	Rhythm is measurable temporal organization with multiple candidate mechanisms and cross-cultural constraint.	Generalize as a DR/Open temporal-organization hypothesis rather than a music metaphor.
Schema / self-reference / fluency	Prior congruence, self-relevance, organization, elaboration, and processing ease can alter memory and felt response.	Treat these as direct rival explanations that resonance must outperform incrementally.
Event cognition	Ongoing activity is segmented into meaningful events; boundaries depend on prediction and conceptual structure and affect memory.	Supplies a non-musical bridge for temporal organization, interruption, transition, and return.
Human-AI feedback	Repeated AI interaction can alter later unaided human judgments.	Supports history-shaped human starting states, not the full resonance/rhythm theory.

articulated; a problem may be epistemically significant before a stable hypothesis is formed.

5.1 Meaning plurality without dilution

The strong thesis is preserved by refusing to equate *meaning* with only one linguistic or propositional form. For analytic purposes, let the reader-conditioned meaning relation be typed as

$$\mu_n = (\mu_n^{aff}, \mu_n^{prag}, \mu_n^{auto}, \mu_n^{conc}, \mu_n^{epi}),$$

where the components denote affective significance, pragmatic/action significance, autobiographical significance, conceptual-semantic significance, and epistemic significance. These are *readout modes*, not biological modules and not an exhaustive ontology. Any subset may be active in a given episode. The unifying claim is stronger, not weaker: **experience is the phenomenon as it becomes significant for the agent through one or more of these mutually interacting meaning relations.**

This typing closes an equivocation without giving up the identity thesis. It explains how preverbal or difficult-to-name experience can still be meaningful, how language can later become constitutive, and why epistemic meaning must remain separable from affective force. In particular,

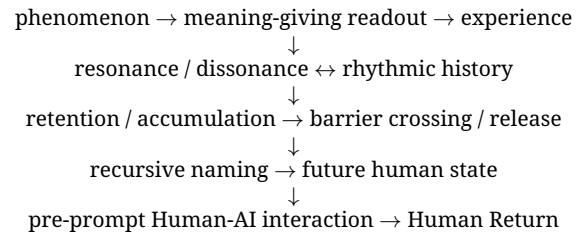
$$\mu^{aff} \neq \mu^{epi}, \quad \mu^{auto} \neq \mu^{epi}, \quad \mu^{prag} \neq \text{truth}.$$

A phenomenon can matter deeply without thereby being correctly interpreted.

5.2 Strong novelty claim: architectural closure

The paper does not claim that phenomenology, enaction, appraisal, schema theory, or event cognition failed to notice significance. Its strong novelty claim is architectural. Across the literature reviewed here, the relevant phenomena are dis-

tributed across separate traditions; the present paper places them into one role-separated sequence with explicit non-collapse and deletion rules:



The contribution is therefore not a synonym for sense-making. It is a closed readout-retention architecture that links lived significance to temporal history, congruence, abrupt transition, Human-AI feedback, and unaided retained capability while keeping truth and warrant external to experiential force.

6. Readout-Native Formalization

The formalization must preserve Readout Genesis order: retained difference first, semantic naming only after a valid domain translation. No psychological noun is promoted to a root primitive [1]. The equations below are **typed dependency maps and candidate domain dynamics**, not already estimated universal psychological laws. Their purpose is to make the strong thesis auditable rather than metaphorical.

Let x_n denote a bounded event trace or encountered pattern made available to the human domain at step n . Let H_n denote the human's retained organization relevant to the current domain, and c_n the context. A finite human readout is

$$r_n = \mathcal{R}_H(x_n \mid H_n, c_n).$$

The readout is not the world event itself. It is what

this finite reader can access under declared conditions.

Meaning-giving is then represented schematically as

$$\mu_n = \Psi_H(r_n, H_n, c_n, Q_n),$$

where μ_n is the reader-conditioned vector of active significance relations under question or criterion Q_n . The scalar notation μ_n is retained when only one declared readout mode is in scope. This is intentionally relational: meaning is not stored inside x_n , and it is not produced independently of x_n .

The MEMK domain already defines experience as

$$E_n = \Phi_E(x_n, \mu_n, \kappa_n, c_n),$$

with the compact interpretation that experience is a non-additive event-trace-meaning configuration under context [2]. The present paper therefore makes explicit a thesis already latent in the programme:

Experience = phenomenon-as-meaningfully-read

within a declared human domain.

The phrase *giving meaning* can now be stated without mystification:

μ_n is the relation that makes x_n significant
for H_n under c_n, Q_n .

The human does not create the root event; the human creates access and significance under bounded conditions.

Naming is a further operator:

$$\ell_n = \mathcal{L}_H(E_n, \mu_n \mid H_n, c_n),$$

where ℓ_n may be a word, description, metaphor, story, question, or formal concept. The architecture permits

$$E_n, \mu_n \text{ before stable } \ell_n,$$

but does not require this order in every case because ℓ_n can feed back into later μ_{n+1} .

Retention is selective:

$$H_{n+1} = U_H(H_n, \text{Retain}(E_n, \mu_n, r_n), \delta_{n:n+1}^{world}).$$

Thus experience is neither identical to memory nor automatically learning.

7. Resonance: External-Internal Experiential Congruence

The present paper corrects the earlier *Meaning Before Naming* use of *resonance*. Resonance is not defined as mere accessibility change and not as physical oscillation. It is the reader-relative condition in which a current externally prompted experience corresponds with a retained internal experiential organization.

Let E_n^{cur} denote the current meaning-given experiential configuration and let

$$I_{H,n} = \text{Retrieve}(\mathcal{M}_{H,n} \mid c_n, Q_n)$$

denote a context-relevant retained experiential organization retrieved from human memory/history. Define a declared congruence diagnostic

$$C_H(E_n^{\text{cur}}, I_{H,n} \mid c_n, Q_n) \in [0, 1].$$

Then experiential resonance is

$$\text{Res}_H(n) = C_H(E_n^{\text{cur}}, I_{H,n} \mid c_n, Q_n).$$

This is a DR/Open domain diagnostic, not a validated psychometric scale.

The intended phenomenology is familiar: a person encounters a melody, sentence, face, scene, ritual, story, or AI response and experiences, “this corresponds to something already inside me,” perhaps before being able to articulate what that correspondence is. The external encounter need not literally contain the internal memory; the internal structure need not be an exact copy of the external pattern. Resonance names **fit across the external-internal experiential boundary**.

Four non-collapses are mandatory:

$$\begin{aligned} \text{Res} &\neq \text{Identity}, & \text{Res} &\neq \text{Truth}, \\ \text{Res} &\neq \text{Retention}, & \text{Res} &\neq \text{Improvement}. \end{aligned}$$

A conspiracy narrative can resonate with retained fear. A prejudicial frame can resonate with prior social experience. A false reassurance can resonate with hope. Congruence explains force for the reader, not warrant.

Strong-form construct criterion. Resonance is retained as a distinct technical object only if an explicit congruence measure contributes incremental held-out prediction after controlling for schema congruence, self-reference, familiarity, processing fluency, appraisal, valence, and autobiographical retrieval. Formally, for an outcome family Y ,

$$\Delta_{\text{Res}} = \mathcal{P}(Y \mid B, \text{Res}) - \mathcal{P}(Y \mid B),$$

where B denotes the preregistered rival set and $\mathcal{P}$ is a declared out-of-sample predictive score. The strong resonance claim earns technical status when $\Delta_{\text{Res}} > 0$ reproducibly under frozen criteria. If not, the phenomenology of fit remains describable, but the special diagnostic should be removed.

Resonance is therefore an interface construct. Candidate mechanisms underneath it may include autobiographical retrieval, schema congruence, prediction, association, affective appraisal, embodied readiness, familiarity, or social framing. If these variables fully explain the relevant outcomes and the congruence construct adds no discriminating value, resonance should be removed as a separate technical object.

8. Rhythm: Structured Recurrence Across Ordered History

Rhythm occupies a different level. It does not mean internal-external congruence. It describes temporal organization of recurrence.

Readout Genesis already supplies ordered lineage before continuous-time semantics [1]. Human Learning includes spacing, retrieval, repeated practice, and workflow; Human LoRA includes rehearsal and selective consolidation; From Problem to Hypothesis models recent-path history; Fusion/Tunnel states that repeated traversal can make a route easier to re-enter. The residual object is therefore not musical rhythm per se but a domain-level temporal structure:

Rhythmic organization is the structured pattern of recurrence, spacing, interruption, emphasis, and return across an ordered lineage, where the organization of prior encounters may alter readiness or accessibility of subsequent readouts.

Let the ordered encounter history be

$$\mathcal{T}_n = \{(x_k, t_k, w_k)\}_{k \leq n},$$

where t_k is order/time and w_k is a declared salience or role weight. A rhythm descriptor

$$\mathcal{R}_n = \Omega(\mathcal{T}_n, \mathcal{B}_n)$$

may summarize recurrence intervals, returns, interruptions, local regularity, and a declared event-boundary sequence $\mathcal{B}_n$. Event-segmentation research makes this extension substantive: boundaries can reorganize prediction, attention, and later memory even when total exposure is similar [30, 31]. This is not yet a causal law of human meaning; it is a stronger temporal hypothesis than exposure count alone.

The semantic-momentum object from *From Problem to Hypothesis* provides one possible downstream effect:

$$m_{t+1}(e) = \rho m_t(e) + \mathbf{1}[e_t = e], \quad 0 \leq \rho < 1,$$

where recent repeated traversal makes the same semantic direction easier to re-enter for a limited history window [6]. The present paper generalizes the question but not the empirical claim: does the temporal organization of recurrence add predictive value beyond mere exposure count and current state?

The following distinctions are therefore required:

order $\neq$ rhythm, repetition $\neq$ rhythm,
rhythm $\neq$ meaning, rhythm $\neq$ retention.

Music is a particularly clear phenomenological probe because timing, pause, repetition, beat, and return are explicit, but the architecture also generates testable questions about learning sequences, therapeutic dialogue, ritual, narrative, and Human-AI conversation.

9. Accumulation, Potential, Barrier Crossing, and Release

The user's motivating intuition is that internal experience can accumulate "power" and then be released explosively when an encounter unlocks it. The programme already contains a disciplined way to retain this intuition without importing physical energy.

MEMK v0.2 defines a domain-level semantic potential $V_{\text{sem}}(z; c, T)$ as an informational stability landscape, an informational work functional W_{info} , and an activation barrier $\Delta V_{\text{old} \rightarrow \text{new}}^\dagger$ separating an old stable meaning basin from a candidate new basin [3]. It explicitly forbids reporting semantic potential or meaning-energy as literal physical energy. It also distinguishes shock-like impulse, coherent sudden insight, gradual learning, and life-changing transformation.

This allows a precise reinterpretation of "stored power." The defensible object is not stored joules but **retained transition readiness**: accumulated history may deepen an old basin, create unresolved conflict, strengthen a latent alternative, or repeatedly perform low-amplitude informational work on the current semantic organization.

For repeated low-amplitude updates, MEMK already proposes

$$\Delta W_{j,N} = \sum_{n=1}^N \eta_n \text{eligible-gradient}_n,$$

and states that a transition may occur after cumulative effective work crosses a barrier or slowly reshapes the stability landscape so the old basin loses stability [3]. The present paper gives this an experiential reading.

A current encounter may contribute effective work according to source strength, reader eligibility, context, and alignment. Resonance can be treated as one possible *alignment term*, not as energy itself:

$$W_n^{\text{eff}} = W_{\text{info},n} g(\text{Res}_H(n), \text{eligibility}, \text{context}),$$

with g left DR/Open. A transition candidate appears when cumulative effective work exceeds a declared barrier:

$$\sum_{k \leq n} W_k^{\text{eff}} \geq \Delta V_{\text{old} \rightarrow \text{new}}^\dagger.$$

The strong claim is not that all psychological change obeys one universal threshold. It is that **some experience trajectories may be better described by a transition topology than by smooth accumulation alone**. This is an empirical wager rather than a retreat. A continuous rival can be written as

$$z_{n+1} = z_n + \alpha W_n^{\text{eff}} + \epsilon_n,$$

while the transition model permits a state-dependent regime change once cumulative work crosses the declared barrier. The barrier claim

gains scientific force only if it predicts abrupt reorganization, hysteresis, delayed settling, or cross-context persistence that the continuous model misses on held-out trajectories.

The phenomenological outcome may be experienced as sudden release: crying, laughter, speech, movement, confession, creative production, insight, decisive action, or a rapid reframing. The paper calls this **transformative release candidate** when the transition is abrupt, while reserving **durable transformation** for change that persists across memory, view, prediction, action, and context.

Hence:

release $\neq$ transformation,
intensity $\neq$ truth,
shock $\neq$ insight,
barrier crossing $\neq$ improvement.

MEMK itself requires persistence, cross-context stability, held-out reuse, and lineage preservation before a life-changing transformation report is warranted [3].

10. Naming as a Recursive Operator

A theory of experience as meaning-giving cannot place language outside the process. Language is both an articulation channel and a potential reorganizer.

Before naming, a person may have an active pattern of felt significance, relational accessibility, action readiness, or mismatch. Naming can make this structure socially inspectable and revisable. But linguistic evidence shows that category words can also alter perception and affective processing [16, 17]. Therefore the architecture is recursive:

$$\mu_n \rightarrow E_n \rightarrow \ell_n \rightarrow \mu_{n+1} \rightarrow E_{n+1}.$$

Language can perform at least four roles: (1) **disclosure**, making a previously tacit relation nameable; (2) **compression**, reducing a high-dimensional lived configuration to a portable label; (3) **restructuring**, changing future distinctions and expectations; and (4) **transport**, exporting a bounded readout to another person or AI system.

This preserves the strongest correction from hostile review: *meaning before naming* cannot mean a fully completed semantic proposition waiting intact for a word. The defensible claim is that **meaning-giving can begin pre-explicitly, while naming can later become constitutive of the next experiential state**.

11. Human-AI Coupling Begins Before the Prompt

The experience theory changes the placement of Human-AI interaction. This paper names the **Pre-Prompt Human State Principle**: *a prompt is not the start of the human trajectory; it is a bounded*

articulation of an already meaning-shaped, history-bearing human state. The principle is architectural, while the size and mechanisms of pre-prompt history effects remain empirical questions.

Let H_t contain retained experiences, concepts, affective weights, values, unresolved differences, and recent interaction history. The human articulates only a partial readout:

$$H_t \xrightarrow{\mathcal{L}_H} Q_t.$$

The AI receives Q_t , not H_t itself. It returns output Y_t , which the human reads through the same embodied and retained organization:

$$Y_t \xrightarrow{\mathcal{R}_H} E_{H,t}^{AI}.$$

That output can resonate with prior experiential organization, fail to resonate, create dissonance, or restructure future meaning-giving. If retained,

$$H_{t+1} = U_H(H_t, E_{H,t}^{AI}, \text{world correction, other experience}).$$

The next question is then produced from a changed state:

$$H_{t+1} \rightarrow Q_{t+1}.$$

This is where rhythm becomes a Human-AI research question. Dialogue has recurrence, turn order, delay, interruption, restatement, and return. A frame repeated across several turns may become easier to re-enter. A strategically spaced counterexample may preserve resistance better than continuous affirmation. A pause that requires unaided retrieval may change retention differently from immediate completion. These are hypotheses, not established design laws.

Glickman and Sharot provide a strong external ally to the general history-shaped claim: repeated exposure to AI judgments can alter later human judgments and amplify bias [32]. Their results do not identify experiential resonance or conversational rhythm as mechanisms. They show that the human state entering a later judgment can differ because of prior AI interaction.

The pre-prompt architecture is therefore:

$$\boxed{H_t \rightarrow Q_t \rightarrow AI_t \rightarrow Y_t \rightarrow E_{H,t} \rightarrow \text{Retain?} \rightarrow H_{t+1} \rightarrow Q_{t+1}}$$

with resonance and rhythm as candidate domain-level modifiers of uptake and re-entry.

12. From Retained Experience to CTSA Human Return

Not every retained experiential change is immediately an explicit capability. CTSA Human-Return Readout asks a narrower question: after AI support is removed, what remains demonstrably usable by the human? Its four classes

are Conceptual, Tool-Selection, Skill-Execution, and Alternative-Generation Return [9].

The present paper therefore inserts a lower layer:

retained experiential reorganization
→ possible later CTSA crystallization.

A resonant encounter may leave only a sensitivity. A rhythmically repeated distinction may become easier to retrieve but still fail a transfer test. A sudden release may produce a new narrative without a new skill. None should be counted as CTSA gain until the unaided human can reconstruct, discriminate, transfer, select, execute, or generate under frozen criteria.

Possible developmental trajectories include

retained sensitivity → C_{return} ,
retained reweighting → A_{return} ,

or

retained reorganization → $Q_{\text{next}}^{\text{better}}$.

These are open pathways, not universal causal sequences.

13. Non-Collapse Laws

The paper's strongest contribution may be the role separation it enforces. The following non-collapses are constitutive:

- phenomenon ≠ experience;
- event trace ≠ meaning;
- meaning-giving ≠ truth-making;
- pre-explicit significance ≠ fully formed proposition;
- naming ≠ passive reporting;
- resonance ≠ identity;
- resonance ≠ truth;
- rhythm ≠ repetition count;
- rhythm ≠ meaning;
- accessibility ≠ warrant;
- semantic attraction ≠ evidence;
- informational work ≠ physical energy;
- shock ≠ coherent insight;
- release ≠ durable transformation;
- retention ≠ improvement;
- Human-AI amplification ≠ synergy;
- coupled performance ≠ Human Return.

These laws block a recurring error: a phenomenon can become experientially powerful because it fits, repeats, returns, or crosses a transition threshold, yet remain false, harmful, manipulable, or epistemically unsupported.

14. Empirical Programme: A Hostile Test, Not a Demonstration

The first empirical programme should try to kill the theory, not illustrate it. Each construct must add out-of-sample explanatory value beyond simpler rivals.

14.1 H1 - Meaning-giving beyond arousal [Open]

Hold stimulus intensity approximately constant while manipulating event relevance to prior goals, autobiographical memory, or learned structure. Measure pre-label discrimination, action tendency, relational accessibility, and later transfer. The claim survives only if the meaning-giving model predicts structured differences beyond undifferentiated arousal.

14.2 H2 - Resonance as external-internal congruence [Open]

Build individualized retained-experience profiles before exposure. Present matched stimuli that are congruent or incongruent with those profiles while controlling familiarity and valence. Model schema congruence, self-reference, processing fluency, appraisal, and autobiographical retrieval explicitly. Test whether declared external-internal congruence predicts immediate felt fit, selective retrieval, later interpretation, and retention beyond this rival set. Resonance retains strong technical status only with reproducible incremental held-out value.

14.3 H3 - Rhythm beyond exposure count [Open]

Hold the number and semantic content of exposures constant while varying temporal organization: massed repetition, spaced recurrence, regular return, irregular return, interruption, delayed re-entry, and event-boundary placement. Measure accessibility, event segmentation, unaided recall, and later transfer. The rhythm construct earns a distinct role only if temporal organization predicts outcomes beyond repetition count, exposure duration, and boundary count alone.

14.4 H4 - Resonance x rhythm interaction [Open]

Cross congruence with temporal organization. A strong prediction is not simply "more repetition gives more effect," but that structured return to a congruent pattern may alter re-entry or retention differently from equally frequent incongruent or temporally disorganized exposure. A null interaction would narrow the integrated theory.

14.5 H5 - Accumulation versus single shock [Open]

Compare (a) repeated low-amplitude meaning-relevant encounters followed by a congruent trigger, (b) a single high-intensity trigger without prior accumulation, and (c) repeated neutral encounters. Test for rapid reorganization, immediate expression, delayed retention, cross-context transfer, and possible hysteresis. Fit both continuous-

accumulation and transition/barrier models before inspecting outcomes. The strong barrier claim survives when the transition model predicts held-out trajectory changes that the continuous rival misses.

14.6 H6 - Naming as recursive intervention [Open]

Measure pre-label organization, then randomly assign free naming, supplied labels, no-label control, or semantic-interference conditions. Test whether later readout differs even when the original stimulus is held constant. Existing language and affect-labeling studies motivate this design but do not establish the full architecture [16, 17].

14.7 H7 - Release is not transformation [Derived guardrail]

Separate immediate intensity or externalization from delayed retention, cross-context stability, action change, and Human Return. A dramatic reaction with no delayed change counts as release without durable transformation.

14.8 H8 - Human-AI pre-prompt state [Open/externally allied]

Randomize repeated AI feedback trajectories while freezing task content and later remove the AI. Measure pre-response human judgments, question quality, CTSA return, and resistance to contrary evidence. Glickman and Sharot show that prior AI interaction can alter later judgments; the proposed test asks whether resonance, rhythm, or meaning-history variables add explanatory value [32].

14.9 H9 - Meaning-mode decomposition [Open]

Test whether the typed meaning relation adds structure beyond a single undifferentiated significance score. Preregister affective, pragmatic, autobiographical, conceptual, and epistemic readouts without treating them as latent biological factors. The identity thesis does not require all five to dissociate psychometrically; it predicts that experience can be significant in more than one way and that collapsing epistemic meaning into affective force will systematically obscure warrant-sensitive outcomes.

14.10 Rival model ladder

Every study should preregister at least five rivals:

M_0 = arousal/intensity only,

M_1 = familiarity/exposure,

M_2 = prediction/surprise,

M_3 = language/category construction,

M_4 = memory/schema retrieval.

with the present readout-retention model M_5 required to add held-out explanatory or predictive value. If it does not, its extra variables should be deleted.

15. Legitimate Defeaters

The architecture should be narrowed or abandoned if well-specified studies show that:

- experience is fully predicted by stimulus features and arousal with no added role for reader history, context, or significance;
- pre-explicit relational measures add no information beyond later verbal labeling;
- external-internal congruence adds no value beyond schema activation, self-reference, familiarity, processing fluency, appraisal, or autobiographical retrieval;
- temporal organization adds no predictive value beyond exposure count, current state, and event-boundary structure;
- repeated low-amplitude history does not alter later transition probability after current-state variables are controlled;
- apparent barrier crossings are fully explained by response threshold artifacts or demand characteristics;
- naming is empirically passive in the relevant domain and does not alter later readout;
- dramatic release does not dissociate from durable transformation under delayed and cross-context measures;
- Human-AI history effects disappear after controlling current prompt content and explicit memory;
- CTSA-style unaided tests reveal no durable capability despite claimed transformation.

The deletion rule is explicit: a construct that performs no distinct explanatory or measurement work should not survive as programme vocabulary.

16. Discussion

The paper began from a simple proposition: *experience is the giving of meaning to phenomena*. After constraint review, that proposition survives only in a specific form. The world is not denied; the reader is not sovereign; and meaning is not treated as a substance. Experience is a relation in which an encountered phenomenon becomes significant for a finite embodied agent. The reader contributes history, expectation, value, memory, and conceptual organization; the phenomenon contributes structure and resistance; context selects which relations become live.

This formulation helps resolve several tensions in the prior programme. First, it explains why *Human Learning* can begin methodologically from lived words without claiming that words are phenomenologically first. A lived word is an inspectable articulation of an already meaning-shaped relation. Second, it strengthens *Human LoRA*: experience is not merely input to learning but the meaning-bearing encounter whose selectively retained consequences

can alter the future reader. Third, it clarifies *Fusion/Tunnel*: dialogue changes not only propositions but the experiential organization from which later propositions are produced. Fourth, it gives CTSA a lower boundary: explicit Human Return may crystallize from retained experiential reorganization without exhausting it.

The corrected definition of resonance is central. Earlier drafts risked using resonance as a loose synonym for accessibility. The present paper instead reserves it for current external-internal experiential congruence. This move makes the construct more precise but also more vulnerable to empirical defeat. It may turn out that familiar schema-congruence or autobiographical retrieval measures do all the needed work. If so, the programme should keep the phenomenon and drop the special label.

Rhythm likewise becomes useful only when stripped of mystique. Its role is structural and temporal. Recurrence, spacing, interruption, and return may organize the conditions under which a relation becomes easier to re-enter. Music makes this structure visible, but the question generalizes to education, ritual, clinical dialogue, social media, and AI conversation. Importantly, repeated ease does not confer truth. A false route can become rhythmically familiar and experientially resonant.

The most provocative extension concerns accumulation and release. Human beings often describe long-held experience as building pressure and then “bursting out.” The programme can preserve this phenomenology without claiming literal stored energy. MEMK’s informational work and semantic-potential barrier offer a typed domain model: repeated low-amplitude updates may accumulate, an aligned encounter may increase effective work, and a transition may become abrupt when the old organization loses stability. The scientific burden is high. The architecture must discriminate gradual accumulation from simple exposure, threshold artifacts, memory retrieval, and post-hoc narrative.

Finally, the Human-AI implication is not that AI secretly controls an invisible emotional field. The stronger and testable claim is that AI enters a meaning-bearing temporal trajectory that already exists before the prompt. It is that Human-AI interaction occurs inside a history-bearing human system. AI outputs are themselves phenomena to be meaningfully read. They can fit or clash with retained experience, recur with particular timing, and become selectively retained. The next prompt is therefore partly an outcome of prior interaction. External empirical evidence already establishes that repeated AI judgments can shift later human judgments [32]; the next research question is what combination of meaning-giving, resonance, temporal organization, resistance, and retention best explains when such influence becomes learning, fixation, or epistemic tunnel.

17. Standalone Thesis

The programme can now be compressed into one statement:

Experience is the meaning-giving relation through which a phenomenon becomes significant for a finite embodied reader. Meaning here is not restricted to propositions: affective, pragmatic, autobiographical, conceptual, and epistemic significance can coexist within one lived configuration. The relation is constrained by the encountered world and shaped by retained history, context, expectation, value, embodiment, and available concepts. When a current externally prompted experience fits retained internal experiential organization, resonance occurs as external-internal congruence. When related encounters recur with temporal structure, rhythm organizes recurrence, spacing, interruption, emphasis, boundary, and return. Repeated retained changes can produce transition readiness; a sufficiently aligned encounter may contribute to barrier crossing and transformative release. Naming can both articulate and restructure experience, retention determines what persists, and independent resistance determines whether what persists deserves epistemic trust. The Pre-Prompt Human State Principle follows: Human-AI interaction begins before the prompt because the prompt is a bounded readout of a previously meaning-shaped and history-bearing state. The quality of collaboration should ultimately be judged by what the human can still discriminate, revise, transfer, and use after the system is gone.

18. Conclusion

The proposed theory does not reduce experience to information processing, language, emotion, or memory. Nor does it place an unconstrained private world between the agent and reality. It treats experience as a bounded relation: phenomenon plus meaning-giving under a finite reader. From that relation, several previously separate programme threads become continuous. Pre-explicit significance explains why naming can lag experience; recursive language explains why naming can later change it; resonance explains why some encounters feel deeply fitted to retained life history; rhythm explains why the temporal organization of return may matter independently of content; retention explains why some encounters disappear while others alter future readout; barrier crossing explains why accumulated change can sometimes become abrupt; and Human Return explains why a powerful interaction still does not count as durable human capability until the AI is gone and the person can act, discriminate, revise, or generate independently.

The strongest conclusion is therefore the strong thesis itself, now made role-separated and testable: **a phenomenon becomes human experience through meaning-giving; meaning-giving is plural in mode, history-shaped, embodied, and world-constrained; resonance is the congruence between present external experience and retained internal experience; rhythm organizes the temporal return of encounters; accumulated retained change can create transition readiness and transformative release; and whatever becomes easier to feel, think, repeat, or retain must still face independent resistance before it earns epistemic status.** In Human-AI systems, this means the epistemic trajectory begins before the prompt and cannot be evaluated only from coupled output.

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References

References

- [1] Readout Genesis Programme. (2026). *READOUT_GENESIS_MEMK_DOMAIN v0.3.0 ANCHORED*. Root-connected semantic-cognitive domain specification, 23 July 2026. Architecture-conditional / unresolved domain.
- [2] Readout Genesis Programme. (2026). *READOUT_GENESIS_MEMK_DOMAIN v0.3.0 ANCHORED*: Meaning-Experience-Memory-Knowledge domain definitions. 23 July 2026.
- [3] Readout Genesis Programme. (2026). *READOUT_GENESIS_MEMK_DOMAIN v0.2.0*: Informational adaptation, semantic potential, informational work, barrier crossing, and transformation protocol. 23 July 2026.
- [4] Lahtee, Y. (2026). *Experience Is the Human LoRA: A Readout-Retention Theory of Selective Model Change*. Open Civil Science Initiative preprint, 18 July 2026.
- [5] Lahtee, Y. (2026). *Human Learning as Epistemic Architecture: A Method for Word Mapping, Life-Concept Graphs, Constraint Testing, and Corrigible Agency in the AI Age*. Revised v4 preprint, 28 June 2026.
- [6] Lahtee, Y. (2026). *From Problem to Hypothesis: Dynamic Semantic Mobility, History-Shaped Access, and the Readout-Discriminable Hypothesis Space*. Standalone final author draft, GLOSA K0, 4 September 2026.
- [7] Lahtee, Y. (2026). *Epistemic Fusion or Epistemic Tunnel: A Phenomenologically Anchored, History-Shaped Architecture of Human-AI Semantic Coupling, Retention, and Epistemic Direction*. Human-AI Readout Programme, Genesis-aligned v7, 5 September 2026.
- [8] Lahtee, Y. (2026). *Meaning Before Naming: A Readout-Retention Architecture of Affective-Semantic Reorganization*. Standalone Concept Note v1.0 FULL, 5 September 2026.
- [9] Lahtee, Y. (2026). *CTSA Human-Return Readout: A Session-Boundary Measurement Architecture for Retained Human Capability After AI*. Epistemic Note v9.1 FINAL, 5 September 2026.
- [10] Smith, D. W. (2013). Phenomenology. In *The Stanford Encyclopedia of Philosophy* (substantive revision 16 December 2013). Stanford University.
- [11] Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press. <https://doi.org/10.7551/mitpress/6730.001.0001>
- [12] Di Paolo, E. A. (2005). Autopoiesis, adaptivity, teleology, agency. *Phenomenology and the Cognitive Sciences*, 4(4), 429–452. <https://doi.org/10.1007/s11097-005-9002-y>
- [13] Thompson, E. (2007). *Mind in Life: Biology, Phenomenology, and the Sciences of Mind*. Harvard University Press.
- [14] Scherer, K. R., & Moors, A. (2019). The emotion process: Event appraisal and component differentiation. *Annual Review of Psychology*, 70, 719–745. <https://doi.org/10.1146/annurev-psych-122216-011854>
- [15] Barrett, L. F., Mesquita, B., Ochsner, K. N., & Gross, J. J. (2007). The experience of emotion. *Annual Review of Psychology*, 58, 373–403. <https://doi.org/10.1146/annurev.psych.58.110405.085709>
- [16] Lindquist, K. A., Barrett, L. F., Bliss-Moreau, E., & Russell, J. A. (2006). Language and the perception of emotion. *Emotion*, 6(1), 125–138. <https://doi.org/10.1037/1528-3542.6.1.125>
- [17] Lieberman, M. D., Eisenberger, N. I., Crockett, M. J., Tom, S. M., Pfeifer, J. H., & Way, B. M. (2007). Putting feelings into words: Affect labeling disrupts amygdala activity in response to affective stimuli. *Psychological Science*, 18(5), 421–428. <https://doi.org/10.1111/j.1467-9280.2007.01916.x>
- [18] Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181–204. <https://doi.org/10.1017/S0140525X12000477>
- [19] Furutachi, S., & Hofer, S. B. (2026). Rethinking predictive processing. *Annual Review of Neuroscience*, 49, 471–494. <https://doi.org/10.1146/annurev-neuro-102124-031410>
- [20] Ghosh, V. E., & Gilboa, A. (2014). What is a memory schema? A historical perspective on current neuroscience literature. *Neuropsychologia*, 53, 104–114. <https://doi.org/10.1016/j.neuropsychologia.2013.11.010>

- [21] Alonso, A., van der Meij, J., Tse, D., & Genzel, L. (2020). Naive to expert: Considering the role of previous knowledge in memory. *Brain and Neuroscience Advances*, 4, 1–17. <https://doi.org/10.1177/2398212820948686>
- [22] Kensinger, E. A., & Ford, J. H. (2020). Retrieval of emotional events from memory. *Annual Review of Psychology*, 71, 251–272. <https://doi.org/10.1146/annurev-psych-010419-051123>
- [23] Beckers, T., & Kindt, M. (2017). Memory reconsolidation interference as an emerging treatment for emotional disorders: Strengths, limitations, challenges, and opportunities. *Annual Review of Clinical Psychology*, 13, 99–121. <https://doi.org/10.1146/annurev-clinpsy-032816-045209>
- [24] Wiley, J., & Danek, A. H. (2024). Restructuring processes and Aha! experiences in insight problem solving. *Nature Reviews Psychology*, 3, 42–55. <https://doi.org/10.1038/s44159-023-00257-x>
- [25] Snyder, J. S., Gordon, R. L., & Hannon, E. E. (2024). Theoretical and empirical advances in understanding musical rhythm, beat and metre. *Nature Reviews Psychology*, 3, 449–462. <https://doi.org/10.1038/s44159-024-00315-y>
- [26] Jacoby, N., Polak, R., Grahn, J. A., et al. (2024). Commonality and variation in mental representations of music revealed by a cross-cultural comparison of rhythm priors in 15 countries. *Nature Human Behaviour*, 8, 846–877. <https://doi.org/10.1038/s41562-023-01800-9>
- [27] van Kesteren, M. T. R., Ruiter, D. J., Fernandez, G., & Henson, R. N. (2012). How schema and novelty augment memory formation. *Trends in Neurosciences*, 35(4), 211–219. <https://doi.org/10.1016/j.tins.2012.02.001>
- [28] Symons, C. S., & Johnson, B. T. (1997). The self-reference effect in memory: A meta-analysis. *Psychological Bulletin*, 121(3), 371–394. <https://doi.org/10.1037/0033-2909.121.3.371>
- [29] Reber, R., Schwarz, N., & Winkielman, P. (2004). Processing fluency and aesthetic pleasure: Is beauty in the perceiver's processing experience? *Personality and Social Psychology Review*, 8(4), 364–382. https://doi.org/10.1207/s15327957pspr0804_3
- [30] Zacks, J. M., Speer, N. K., Swallow, K. M., Braver, T. S., & Reynolds, J. R. (2007). Event perception: A mind-brain perspective. *Psychological Bulletin*, 133(2), 273–293. <https://doi.org/10.1037/0033-2909.133.2.273>
- [31] Zacks, J. M., & Swallow, K. M. (2007). Event segmentation. *Current Directions in Psychological Science*, 16(2), 80–84. <https://doi.org/10.1111/j.1467-8721.2007.00480.x>
- [32] Glickman, M., & Sharot, T. (2025). How human–AI feedback loops alter human perceptual, emotional and social judgements. *Nature Human Behaviour*, 9, 345–359. <https://doi.org/10.1038/s41562-024-02077-2>